MIL-F-43581B
9 May 1985
SUPERSEDING
MIL-F-43581A
22 September 1969

MILITARY SPECIFICATION

FILM, COPYING, THERMOGRAPHIC (PROJECTION TRANSPARENCIES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This document covers thermographic copying film, which produces a positive, direct reading image from printed, pen or pencil originals when processed through an infra-red copying machine (see 6.1, 6.3 and 6.5).

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

A-A-113	-	Tape, Pressure-Sensitive Film, Office Use
UU-P-121	-	Paper, Bond and Writing, White and Colored
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7530

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:45Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Hazardous Substances Labeling Act

(Application for copies should be addressed to the U.S. Department of Health and Human Services, Food and Drug Administration, (HHS), 5600 Fishers Lane, Rockville, MD 20857.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY PUBLICATIONS

T 411 os-44 Thickness and Density of Paper

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:45Z
Check the source to verify that this is the currentversion before use.

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Materials. The copying film shall consist of a single sheet of transparent film coated with a flexible thermosensitive material conforming to the physical and performance requirements of this document.

3.2 Hazardous materials. The copying film shall comply with Federal Hazardous Substances Labeling Act and shall have no adverse effect on the health of the personnel when used for the intended purpose (see 6.4).

* 3.3 Color. The copying film shall be clean (water-white) or slightly off-white provided the color is not imparted to the projected image.

* 3.4 Size and thickness. The size of the copying film shall be 8-1/2 ± 1/16 by 10-1/2 ± 1/16 inches. The copying film shall have a thickness of 0.001 to 0.007 inch.

3.5 Projection operating position. The operating position shall be easily identified as described in 3.11 (Instructions for use).

* 3.6 Projection performance. The copying film shall produce direct reading, positive images. The projected images shall be sharp, black or near black, uniform in intensity, legible, and easily read. The imaged copying film shall lay flat on the projector stage and shall not curl or wrinkle before or during viewing to the extent that any portion of the projected image is out of focus when tested as specified in 4.2.5.

3.7 Heat stability. The imaged copying film shall not curl, wrinkle or change dimensionally when heated as specified, 4.2.5. The copying film may darken slightly in color, however the transmission density of the background must not increase more than 0.10 units, when compared to a control and there shall be no decrease in the sharpness and legibility of the image when tested as specified in 4.2.5.

* 3.8 Image life on projector. The projected image of the copying film shall be sharp, uniform in intensity, and shall be legible and easily read, when compared to a control. The film shall also lay flat on the overhead projector and shall not curl or wrinkle. The background transmission density for copying film must increase no more than 0.05 units and the transmission image density must decrease no more than 10 percent when tested as specified in 4.2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:45Z
Check the source to verify that this is the current version before use.

* 3.9 Image life under conditions of 105°F and 90 percent relative humidity. The projected image of the copying film shall be sharp, uniform in intensity and shall be legible and easily read as compared to a control. The background transmission density for copying film must increase no more than 0.05 units and the transmission image density must decrease no more than 10 percent when tested as specified in 4.2.5.

* 3.10 Image capability after exposure to 105°F and 90 percent relative humidity. The oven heated unimaged copying film when imaged in accordance with 4.3.4.2, shall produce images that are sharp, uniform in intensity and shall be legible and easily read as compared to a control. The background transmission density for copying film must increase no more than 0.10 units and the transmission image density must decrease no more than 15 percent when tested as specified in 4.2.5.

* 3.11 Instructions for use. The manufacturer shall furnish complete and concise instructions for the use and storage of the copying film and imaged copying film.

* 3.12 Coating. The coating for copying film shall not be removed when tested as specified in 4.2.5.

3.13 Workmanship. The copying film shall be smooth and uniform to touch and appearance. The coating shall be uniformly applied. The base film and coating shall be free from foreign matter, holes, tears, wrinkles, creases and folds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless otherwise disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:45Z
Check the source to verify that this is the current version before use.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item visual examination. The end item shall be examined on both sides for the defects listed below. The lot size shall be expressed in units of 100 sheets. The sample unit shall be one sheet of film. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Appearance	Color, finish and formation not as specified. Not clean; any discoloration, stain or mark.
Construction and workmanship	Presence of foreign matter, hole, tear, crease, wrinkles or folds. Coating not uniformly applied; non-uniform appearance of coating.

4.2.3 End item dimensional examination. The end item shall be examined for dimensions specified in 3.4. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of 100 sheets. The sample unit shall be one sheet of film. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.2.4 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with table I. The sample unit shall be one shipping container fully packaged. The lot size shall be expressed in units of shipping containers. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:45Z
Check the source to verify that this is the current version before use.

TABLE I. Packaging defects

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, or improper taping. Bulged or distorted container.

4.2.5 End item testing. The end item shall be tested for the characteristics listed in table II. The sample unit shall be five sheets of film. The lot shall be unacceptable if any sample unit fails to meet any test requirement specified. The sample size shall be in accordance with the following:

Lot size (units of 100 sheets)	Sample size (no. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Hazardous materials	3.2	1/
Thickness	3.4	4.3.1
Projection performance	3.6	4.3.2
Heat stability	3.7	4.3.3
Image life on projector	3.8	4.3.4
Image life under conditions of 105°F and 90 percent relative humidity	3.9	4.3.5
Image capability after exposure to 105°F and 90 percent relative humidity	3.10	4.3.6
Coating adhesive	3.12	4.3.7

1/ Certificate of compliance. The contractor shall certify that his product has been and is currently commercially in use with no ill effects on personnel handling the product under normal operating conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:45Z
Check the source to verify that this is the current version before use.

4.3 Methods of inspection.

4.3.1 Thickness test. The thickness shall be determined according to TAPPI Method T-411 os-44, except that the test specimen shall consist of not less than five sheets and two thickness determinations shall be made on each sheet. The results shall be averaged and reported to the nearest 0.0001 inch.

4.3.2 Projection performance.

4.3.2.1 Equipment. Infra-red copying machine (see 6.5), overhead projector, which gives a minimum of 2,000 lumens measured on a 60 by 60 inch test target, and viewing screen (matte finished).

4.3.2.2 Procedure. Remove the first page of this document and use as the original. Prepare an imaged transparency in accordance with the instructions provided by the manufacturer of the copying film. View the projected image with the overhead projector positioned about 6 feet from the viewing screen. Standing behind the projector, observe the sharpness, the color and intensity, uniformity, legibility, and ease of reading of the image over the entire projected area. A transparent plate, a mount, or other means shall not be used to hold the imaged transparency in a flat position on the stage of the projector.

4.3.3 Heat stability test.

4.3.3.1 Equipment. Diffuse transmission densitometer equipped with a Wratten 106 W filter or equivalent (Macbeth Model TD 504 or equivalent) (see 6.6).

4.3.3.2 Procedure. Cut the imaged transparency prepared as outlined in the projection performance test into two equal parts. Place one half of the transparency in an oven at 130°F ± 2°F for 4 hours. At the end if this time, remove the specimen from the oven, place on a flat surface, and cool to room temperature. Compare the heat-treated half with the unheated half for curling, wrinkling, dimensions, and color. Measure the diffuse transmission density of the background area of both sheets by using a diffuse transmission densitometer equipped with a Wratten 106 W filter or equivalent (Macbeth Model TD 504 or equivalent). Compare the projected images for sharpness and legibility.

4.3.4 Test for image life on the projector.

4.3.4.1 Equipment. Infra-red copying machine (see 6.5), overhead projector, which gives a minimum of 2,000 lumens measured on a 60 by 60 test target, and viewing screen (matte finished), diffuse transmission densitometer equipped with a Wratten 106 W filter or equivalent (Macbeth Model TD 504 or equivalent).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:45Z
Check the source to verify that this is the current version before use.

4.3.4.2 Procedure. Remove the first page of this document and use as the original. Prepare an imaged transparency in accordance with the instructions provided by the manufacturer. Place the imaged copying film on the projector, with half of the film covered by a sheet of white bond paper conforming to type III of UU-P-121, and allow the imaged copying film to be illuminated for a period of 6 hours. The imaged copying film shall lay flat on the projector and shall not curl or wrinkle. By means of a diffuse transmission densitometer, measure transmission density at points in the image and in the background area that were located approximately 2 inches from the center of the stage. Measurements should be taken on the unshielded surface and the surface under the bond paper which was used as a revelation shield during the test. A transparent plate, a mount, or other means shall not be used to hold the imaged transparency in a flat position on the stage of the projector. The projected image shall be sharp, uniform in intensity, legible, and easily read as compared to the control.

4.3.5 Test for image life under conditions of 105°F and 90 percent relative humidity.

4.3.5.1 Equipment. Infra-red copying machine (see 6.5). Diffuse transmission densitometer equipped with a Wratten 106 W filter or equivalent (Macbeth Model TD 504 or equivalent). Controlled temperature and humidity oven (temperature should be controlled to ± 2°F and humidity to ± 3 percent). Overhead projector, which gives a minimum of 2,000 lumens measured on a 60 by 60 inch test target, and viewing screen (matte finished).

4.3.5.2 Procedure. Prepare an imaged transparency as specified in 4.3.4.2. Hang two sheets of the imaged copying film individually in a controlled temperature and humidity oven for a period of one week. Remove the samples and observe for curling and wrinkling. Measure the diffuse transmission density in the background area and image density by means of diffuse transmission densitometer equipped with a Wratten 106 W filter or equivalent (Macbeth Model TD 504 or equivalent). View the projected image with the overhead projector positioned about 6 feet from the viewing screen. Standing behind the projector, observe the sharpness, intensity, uniformity, legibility, and ease of reading the image over the entire projected area as compared to the control. A transparent plate, a mount or other means shall not be used to hold the imaged transparency in a flat position on the stage of the projector.

4.3.6 Testing for image capability after 105°F and 90 percent relative humidity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:45Z
Check the source to verify that this is the current version before use.

4.3.6.1 Equipment. Infra-red copying machine (see 6.5) controlled temperature and humidity oven, overhead projector which gives a minimum 2,000 lumens measured on a 60 by 60 inch test target, and viewing screen (matte finished), diffuse transmission densitometer equipped with a Wratten 106 W filter or equivalent (Macbeth Model TD 504 or equivalent).

4.3.6.2 Procedure. Hang two sheets of the imaged copying film in a controlled temperature and humidity oven for a period of one week. Remove the unimaged copying film, and observe for curling and wrinkling and prepare two imaged transparencies as specified in 4.3.4.2. View the projected image with the overhead projector positioned about 6 feet from the viewing screen. Standing behind the projector, observe the sharpness, intensity, uniformity, legibility, and ease of reading the image over the entire projected area. Measure the image transmission density and background transmission density by means of diffuse transmission densitometer equipped with a Wratten 106 W filter or equivalent (Macbeth Model TD 504 or equivalent). Compare to the imaged control. A transparent plate, a mount, or other means shall not be used to hold the imaged transparency in a flat position on the stage of the projector.

4.3.7 Coating adhesion test. Adhere a 3 by 3/4 inch piece of tape conforming to type I, class A, of A-A-113 to the coated side of the film. Apply the tape with the long dimension perpendicular to an edge and with approximately 1/2 inch extending past the edge. Hold the film firmly against a flat surface and remove the tape with a sudden jerk using the tape extending past the edge as a pull tab.

5. PACKAGING

5.1 Preservation. Preservation shall be Commercial.

5.1.1 Commercial. Sheets of copying film shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. One thousand sheets of copying film, of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing. One thousand sheets of copying film, of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:45Z
Check the source to verify that this is the currentversion before use.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification.

5.2.3 Commercial packing. Copying film, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), copying film, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, interior packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Special marking. The following special marking shall be legibly printed or stamped in black capital letters on each interior package and shipping container.

"STORE IN A COOL DRY PLACE - USE BEFORE*

* Expiration date of the copying film.

6. NOTES

6.1 Intended use. The copying film is intended for the preparation of direct reading, positive images for use with overhead projectors. These images are prepared from printed, pen or pencil originals by passing the copying film and the original through a commercial copy machine employing dry heat (infra-red) process.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Selection of applicable level of packing (see 5.2).
c. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
d. When palletization is required (see 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:45Z
Check the source to verify that this is the current version before use.

6.3 Printing and writing materials. With the copying film, direct reading positive transparencies may be prepared from markings made with printing and writing materials such as black printers ink, India ink, carbon-containing typewriter ribbon, black sign pens, graphite pencil and carbon containing sign pens.

6.4 Toxicity. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

6.5 Infra-red copying machine. An infra-red copying machine similar to a Thermo-Fax Brand Copy Machine from Minnesota Mining and Manufacturing Company, P. O. Box 3800, St. Paul, MN 55119, shall be used.

6.6 Macbeth Instrument. Macbeth Model TD 504 may be obtained from Macbeth Instrument Corporation, P.O. Box 950, Newburgh, NY.

6.7 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 99

Preparing activity:

Army - GL

Project No. 7530-0197

Review activities:

Army - MD
Air Force - 84

User activity:

Navy - SA

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:45Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:45Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY
Army Natick Research and Development Center
Natick, Massachusetts 01760

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Center
ATTN: STRNC-ES
Natick, MA 01760

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-43581B	Film, Coping, Thermographic (Projection Transparencies)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

Source: https://assist.dla.mil -- Downloaded: 20260-21T21:45Z
Check the source to verify that this is the current version before use.